

Computer Science & IT Engineering

DPP 5 ; Top 250 Questions of DBMS Part- 5



DBMS : DPP 5

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Consider the following FD set F on a relation scheme $R(ABCDEFGH)$
 $A \rightarrow BCD$, $B \rightarrow D$, $BC \rightarrow DEF$, $D \rightarrow A$, $F \rightarrow GH$.

Which of the following statements is/are true for the relation R & FD set F ?

- A** There are 3 candidate keys for relation R .
- B** The set of FD's : $[A \rightarrow B, A \rightarrow C, B \rightarrow D, B \rightarrow E, B \rightarrow F, D \rightarrow A, F \rightarrow H, F \rightarrow G]$ is minimal cover for F .
- C** B is an extraneous Attribute in $BC \rightarrow DEF$.
- D** FD $A \rightarrow H$ is logically implied in FD set F .

$R(ABCDEFGH) \quad A \rightarrow BCD, B \rightarrow D, BC \rightarrow DEF, D \rightarrow A, F \rightarrow GH.$



$R(ABCDEFGH) \quad A \rightarrow BCD, B \rightarrow D, BC \rightarrow DEF, D \rightarrow A, F \rightarrow GH.$



[MCQ]



Consider a relation STUDENT (Name, Subject, Location, Marks).
Student is decomposed into following

Name	Subject	Location	Marks
Madhav	Operating System	Noida	96
Madhav	DBMS	Noida	100

1. $STU_{Sub\ 1}$ (Name, Subject, Location) and $STU_{Sub\ 2}$ (Name, Location, Marks).
2. $STU_{Sub\ 1}$ (Name, Location) and $STU_{Sub\ 2}$ (Subject, Marks).

Which of the following is True?

- A. 1 is lossy but 2 is lossless
- B. 1 is lossless but 2 is lossy
- C. Both 1 and 2 are lossless
- D. Both 1 and 2 are lossy



Name	Subject	Location	Marks
Madhav	Operating System	Noida	96
Madhav	DBMS	Noida	100

1. $STU_{Sub\ 1}$ (Name, Subject, Location) and $STU_{Sub\ 2}$ (Name, Location, Marks).
2. $STU_{Sub\ 1}$ (Name, Location) and $STU_{Sub\ 2}$ (Subject, Marks).

[MCQ]



Assume a relation $R = (P \ Q \ R \ S \ T \ U)$ and functional dependencies: $F = \{PQ \rightarrow RU, RT \rightarrow Q, U \rightarrow S\}$, consider the following two statements:

S_1 : The decomposition of R into $PQRT$ and $PQSTU$ is a dependency preserving decomposition.

S_2 : The decomposition of R into $PQRT$ and $PQSTU$ is a lossless decomposition.

Which of the statement is/are TRUE?

A.

Only S_1 is true.

B.

Only S_2 is true.

C.

Both S_1 and S_2 are true.

D.

Both S_1 and S_2 are false.

Normal Forms





In 2NF can we have partial Dependency ?





In 3NF can we have Transitive Dependency ?



Normal Forms



15.3.5 Second Normal Form

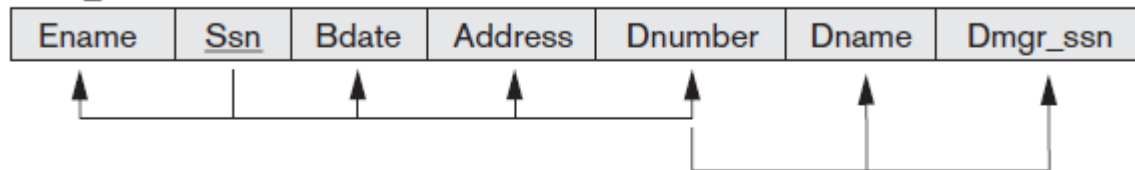
Second normal form (2NF) is based on the concept of *full functional dependency*. A functional dependency $X \rightarrow Y$ is a **full functional dependency** if removal of any attribute A from X means that the dependency does not hold any more; that is, for any attribute $A \in X$, $(X - \{A\})$ does *not* functionally determine Y . A functional dependency $X \rightarrow Y$ is a **partial dependency** if some attribute $A \in X$ can be removed from X and the dependency still holds; that is, for some $A \in X$, $(X - \{A\}) \rightarrow Y$. In Figure 15.3(b), $\{\text{Ssn}, \text{Pnumber}\} \rightarrow \text{Hours}$ is a full dependency (neither $\text{Ssn} \rightarrow \text{Hours}$ nor $\text{Pnumber} \rightarrow \text{Hours}$ holds). However, the dependency $\{\text{Ssn}, \text{Pnumber}\} \rightarrow \text{Ename}$ is partial because $\text{Ssn} \rightarrow \text{Ename}$ holds.

Definition. A relation schema R is in 2NF if every nonprime attribute A in R is *fully functionally dependent* on the primary key of R .

Figure 15.3

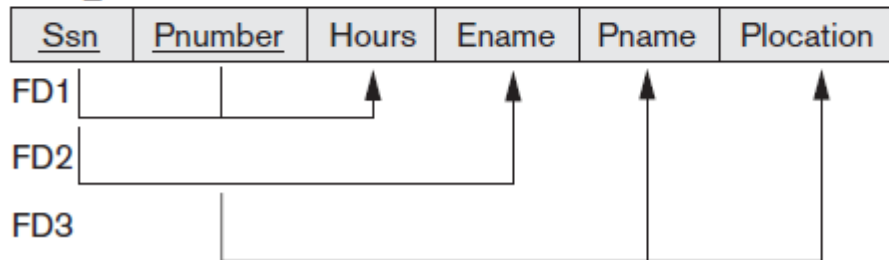
(a)

EMP_DEPT



(b)

EMP_PROJ



Normal Forms

8.17 A functional dependency $\alpha \rightarrow \beta$ is called a **partial dependency** if there is a proper subset γ of α such that $\gamma \rightarrow \beta$. We say that β is *partially dependent* on α . A relation schema R is in **second normal form** (2NF) if each attribute A in R meets one of the following criteria:

- It appears in a candidate key.
- It is not partially dependent on a candidate key.

Normal Forms

Second Normal Form

Definition: A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.

EMP_PROJ

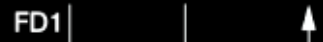
<u>Ssn</u>	<u>Pnumber</u>	Hours	Ename	Pname	Plocation
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2NF Normalization

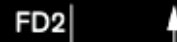
EP1

<u>Ssn</u>	<u>Pnumber</u>	Hours
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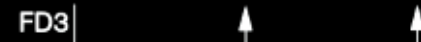
EP2

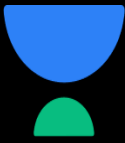
<u>Ssn</u>	Ename
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EP3

<u>Pnumber</u>	Pname	Plocation
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Let $R(A, B, C, D, E, P, G)$ be a relational schema in which the following functional dependencies are known to hold:

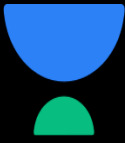
$AB \rightarrow CD$, $DE \rightarrow P$, $C \rightarrow E$, $P \rightarrow C$ and $B \rightarrow G$.

The relational schema R is

- A** In BCNF
- B** In 3NF, but not in BCNF
- C** In 2NF, but not in 3NF
- D** Not in 2NF



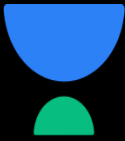
Normal Forms



Third Normal Form

Definition: According to Codd's original definition, a relation schema R is in 3NF if it satisfies 2NF and no nonprime attribute of R is transitively dependent on the primary key.

Definition: A relation schema R is in third normal form (3NF) if, whenever a nontrivial functional dependency $X \rightarrow A$ holds in R either (a) X is a superkey of R , or (b) A is a prime attribute of R .



EMP_DEPT

Ename	<u>Ssn</u>	Bdate	Address	Dnumber	Dname	Dmgr_ssn
↑		↑	↑	↑	↑	↑

3NF Normalization

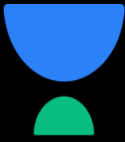
ED1

Ename	<u>Ssn</u>	Bdate	Address	Dnumber
↑		↑	↑	↑

ED2

<u>Dnumber</u>	Dname	Dmgr_ssn
↑		↑

Boyce – Codd Normal Form



Definition: A relation schema R is in BCNF if whenever a nontrivial functional dependency $X \rightarrow A$ holds in R , then X is a superkey of R .



Q.1



$R(ABCDEF) \{AB \rightarrow CD, A \rightarrow C, B \rightarrow EF\}$ Identify the Valid FD's?

- (i) $AB \rightarrow C$
- (ii) $A \rightarrow C$
- (iii) $B \rightarrow D$
- (iv) $AB \rightarrow F$
- (v) $AB \rightarrow D$
- (vi) $C \rightarrow A$
- (vii) $A \rightarrow D$
- (viii) $CD \rightarrow A$

Q.2



$R(ABCDEF) \{ABC \rightarrow DE, DE \rightarrow ABC, AB \rightarrow D, DE \rightarrow F, E \rightarrow C\}$

Identify the Valid FD's?

(i) $ABC \rightarrow D$

(ii) $AE \rightarrow C$

(iii) $AF \rightarrow D$

(iv) $AB \rightarrow D$

(v) $AC \rightarrow D$

(vi) $BC \rightarrow D$

(vii) $DE \rightarrow C$

(viii) $AB \rightarrow F$

(ix) $A \rightarrow F$

Q.3



$R(ABCDEF) \{ABC \rightarrow DE, DE \rightarrow ABC, AB \rightarrow D, DE \rightarrow F, E \rightarrow C\}$

Identify the Partial FD's?

(i) $ABC \rightarrow D$

(ii) $AE \rightarrow C$

(iii) $AF \rightarrow D$

(iv) $AB \rightarrow D$

(v) $AC \rightarrow D$

(vi) $BC \rightarrow D$

(vii) $DE \rightarrow C$

(viii) $AB \rightarrow F$

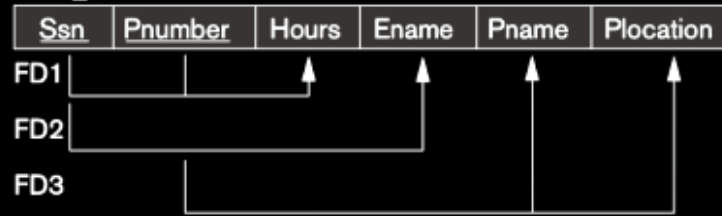
(ix) $A \rightarrow F$

Normal Forms

Second Normal Form

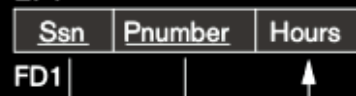
Definition: A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.

EMP_PROJ

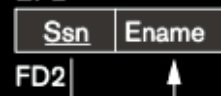


2NF Normalization

EP1



EP2



EP3



Q.4

R(ABCDE) { $AB \rightarrow D$, $C \rightarrow B$, $E \rightarrow A$ } Identify the Partial FD's & Checking 2NF?



Q.5

R(ABCDE) {AB $\rightarrow$ CDE, B $\rightarrow$ C} Identify the Partial FD's & Checking 2NF?



Q.6

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, B \rightarrow D\}$ Identify the 2NF?



Q.7

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$ Identify the 2NF?



Q.8

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$ Identify $AB \rightarrow C$ is the Transitive FD & R is in 3NF or Not ?



Q.9

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$ Show Partial FD may be Transitive or Not ?





Q.10

$R(ABC) \{A \rightarrow B, B \rightarrow C\}$ Check $A \rightarrow C$ Partial FD & Transitive or Not?



[MCQ]

Consider a relation R (P, Q, R, S, T) with the functional dependency's as :

{

$PQR \rightarrow ST$

$S \rightarrow PQ$

}

Which of the following is the total number of superkeys for the above relation R?

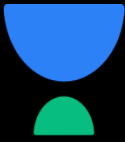
(a) 10

(b) 12

(c) 8

(d) 6

[NAT]

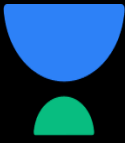


Consider the following relational schema $R(ABCDE)$ with functional dependencies:

$\{AB \rightarrow C, D \rightarrow E, E \rightarrow A, D \rightarrow B\}$

The number of additional relations are required to convert it into lossless join & Dependency Presenting 2NF Decomposition is-

Video Solution of DPP 5



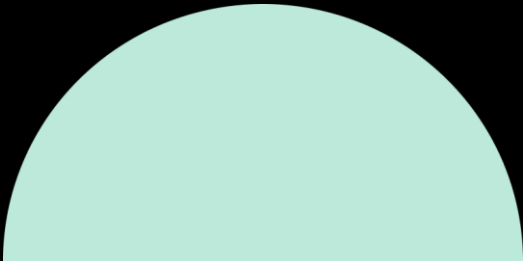
Link: <https://unacademy.com/a/Top-250-Questions-on-DBMS.html%27>

Video Solution of DPP 5 LINK



THANK YOU!

Here's to a cracking journey ahead!

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